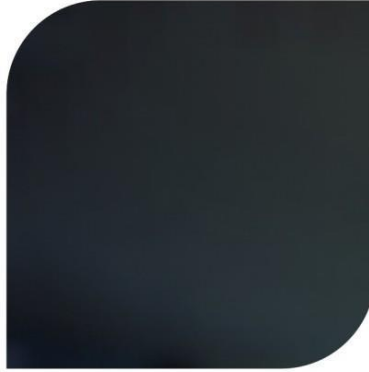
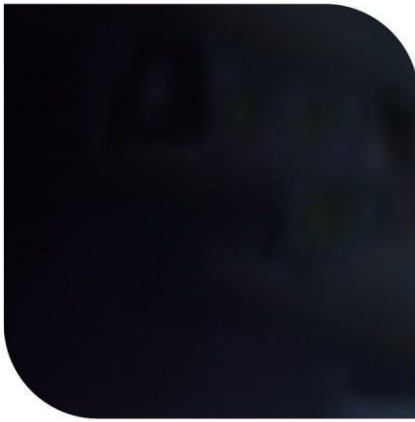


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Real Time Bus Ticket Booking Management System Project Report

A report Submitted in fulfillment of the requirements for the organization



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Abstract

The Real Time Bus Ticket Booking Management System is a Python-based console application developed to simplify and automate the process of bus ticket booking and passenger management. The system allows users to view available buses, check seat availability, reserve tickets, cancel bookings, search passenger details, and generate booking summaries efficiently. It provides a real-time seat tracking mechanism that helps avoid duplicate bookings and improves booking accuracy.

The application includes multiple bus routes, seat layout visualization, ticket generation, booking history, revenue calculation, and passenger record management features. The project is designed using Python concepts such as functions, dictionaries, loops, conditional statements, file-like data handling, and modular programming techniques. This system reduces manual work, improves booking management efficiency, and demonstrates the practical implementation of a real-world transportation reservation system using Python.

Introduction

In today's fast-moving world, transportation systems play an important role in connecting people between cities and states. With the increasing number of passengers travelling daily, managing bus reservations manually has become difficult and inefficient. Traditional ticket booking methods often lead to problems such as long waiting times, duplicate seat bookings, improper record maintenance, and lack of real-time seat availability information. To overcome these issues, computerized booking systems are widely used in the transportation industry.

The **Real Time Bus Ticket Booking Management System** is developed as a smart and efficient solution for handling bus reservation activities digitally. This system helps passengers and administrators manage ticket booking operations quickly and accurately. The application provides features such as viewing available buses, checking routes and departure timings, selecting seats, booking tickets, cancelling reservations, searching passenger records, and monitoring available seats in real time.

The project is designed to simulate a real-world online bus reservation platform where multiple buses operate on different routes with separate seat allocations and fare structures. The real-time seat tracking feature ensures that once a seat is booked, it cannot be reserved again by another passenger. This improves booking accuracy and avoids reservation conflicts. The system also generates booking summaries and revenue reports which help in efficient management of transportation services.

The application is developed using Python programming language with a menu-driven console interface. Various Python concepts such as functions, loops, dictionaries, conditional statements, exception handling, lists, and modular programming are used in the project implementation. The simple structure of the program makes it easy to understand, user-friendly, and suitable for beginners as well as academic mini projects.

The primary aim of this project is to provide an automated and reliable bus reservation system that reduces

manual work and improves operational efficiency. It also demonstrates the practical application of programming in solving real-life management problems related to public transportation systems.

The system is designed in such a way that it provides a smooth and interactive booking experience for users. Passengers can easily choose their preferred bus, view seat layouts, and reserve seats according to availability. The application maintains passenger details securely within the program and updates booking information instantly whenever a ticket is booked or cancelled. This real-time updating mechanism improves reliability and ensures better management of transportation services.

In addition to ticket reservation, the project also focuses on management and monitoring functionalities. Features such as booking history, passenger search, total revenue calculation, and bus summary reports help administrators analyze booking activities effectively. The project not only enhances programming knowledge but also provides practical exposure to developing real-time management systems used in industries such as transportation, travel, and tourism.

Concepts Used

- Python Functions
- Dictionaries
- Nested Dictionaries
- Loops
- Conditional Statements
- Exception Handling
- User Defined Functions
- Menu Driven Programming
- Real Time Seat Tracking
- Data Management
- Input and Output Operations
- String Handling
- List Operations
- Modular Programming
- Console Based Interface
- Passenger Record Management
- Seat Allocation Logic
- Revenue Calculation
- Booking History Management
- Date and Time Module

Key Features

- Multiple Bus Management
- Real Time Seat Availability Tracking
- Online Ticket Booking Simulation
- Ticket Cancellation Facility
- Passenger Search System
- Seat Layout Visualization
- Automatic Ticket Generation
- Booking History Tracking
- Revenue and Collection Report
- Bus Summary Report
- Multiple Destination Support
- Passenger Information Management
- Duplicate Booking Prevention
- User Friendly Menu Interface
- Real Time Data Updating
- Bus Route and Timing Display
- Fare Management System
- Available and Booked Seat Counter
- Error Handling for Invalid Inputs
- Console Based Reservation System

Project Name: Real Time Bus Ticket Booking Management System

Source Code

```
bus.py ×
bus.py > ...
1  # =====
2  # # Real Time Bus Ticket Booking Management System
3  # =====
4  import os
5  from datetime import datetime
6  buses = {
7      1: {
8          "Bus": "Orange Travels",
9          "Route": "Bangalore -> Hyderabad",
10         "Time": "09:00 PM",
11         "Fare": 850,
12         "Type": "AC Sleeper",
13         "Seats": {i: None for i in range(1, 21)}
14     },
15     2: {
16         "Bus": "VRL Travels",
17         "Route": "Bangalore -> Chennai",
18         "Time": "10:30 PM",
19         "Fare": 650,
20         "Type": "Non AC Seater",
21         "Seats": {i: None for i in range(1, 21)}
22     },
23     3: {
24         "Bus": "SRS Travels",
25         "Route": "Bangalore -> Mumbai",
26         "Time": "07:00 PM",
27         "Fare": 1200,
28         "Type": "Volvo Multi Axle",
29         "Seats": {i: None for i in range(1, 21)}
30     },
31     4: {
32         "Bus": "KPN Travels",
33         "Route": "Bangalore -> Goa",
34         "Time": "08:15 PM",
35         "Fare": 950,
36         "Type": "AC Sleeper",
37         "Seats": {i: None for i in range(1, 21)}
38     }
39 }
40 booking_history = []
41 def clear():
42     os.system("cls" if os.name == "nt" else "clear")
43 def line():
44     print("=" * 70)
```

```

45 def available(bus_id):
46     return sum(1 for s in buses[bus_id]["Seats"].values() if s is None)
47 def booked(bus_id):
48     return sum(1 for s in buses[bus_id]["Seats"].values() if s is not None)
49 def total_collection():
50     total = 0
51     for i in buses:
52         total += booked(i) * buses[i]["Fare"]
53     return total
54 def show_buses():
55     line()
56     print("                ONLINE BUS BOOKING SYSTEM")
57     line()
58     for i, b in buses.items():
59         print(f"{i}. {b['Bus']}")
60         print(f"    Route      : {b['Route']}")
61         print(f"    Bus Type   : {b['Type']}")
62         print(f"    Departure  : {b['Time']}")
63         print(f"    Fare       : ₹{b['Fare']}")
64         print(f"    Available  : {available(i)} Seats")
65         print("-" * 70)
66 def seat_layout(bus_id):
67     line()
68     print("SEAT LAYOUT")
69     line()
70     for i in range(1, 21):
71         if buses[bus_id]["Seats"][i]:
72             print(f"[{i:02d}:X]", end=" ")
73         else:
74             print(f"[{i:02d}:A]", end=" ")
75         if i % 4 == 0:
76             print()
77     print("\nA = Available | X = Booked")
78     line()
79 def generate_ticket(bus_id, seat, data):
80     line()
81     print("                BUS TICKET")
82     line()
83     print("Booking Time :", datetime.now().strftime("%d-%m-%Y %I:%M %p"))
84     print("Passenger      :", data["Name"])
85     print("Age            :", data["Age"])
86     print("Gender         :", data["Gender"])
87     print("Phone          :", data["Phone"])
88     print("Bus Name       :", buses[bus_id]["Bus"])
89     print("Route          :", buses[bus_id]["Route"])
90     print("Bus Type       :", buses[bus_id]["Type"])
91     print("Departure      :", buses[bus_id]["Time"])
92     print("Seat Number    :", seat)
93     print("Fare           : ₹", buses[bus_id]["Fare"])
94     line()
95 def book_ticket():
96     show_buses()
97     try:
98         bus_id = int(input("Select Bus Number: "))

```

```

99         if bus_id not in buses:
100             print("Invalid Bus Selection!")
101             return
102         seat_layout(bus_id)
103         seat = int(input("Enter Seat Number: "))
104         if seat not in buses[bus_id]["Seats"]:
105             print("Invalid Seat Number!")
106         elif buses[bus_id]["Seats"][seat]:
107             print("Seat Already Booked!")
108         else:
109             name = input("Enter Passenger Name : ")
110             age = input("Enter Age           : ")
111             gender = input("Enter Gender       : ")
112             phone = input("Enter Mobile Number : ")
113             data = {
114                 "Name": name,
115                 "Age": age,
116                 "Gender": gender,
117                 "Phone": phone
118             }
119             buses[bus_id]["Seats"][seat] = data
120             booking_history.append({
121                 "Bus": buses[bus_id]["Bus"],
122                 "Passenger": name,
123                 "Seat": seat
124             })
125             clear()
126             print("Ticket Booked Successfully!\n")
127             generate_ticket(bus_id, seat, data)
128         except:
129             print("Invalid Input!")
130     def cancel_ticket():
131         show_buses()
132         try:
133             bus_id = int(input("Select Bus Number: "))
134             seat = int(input("Enter Seat Number: "))
135             if buses[bus_id]["Seats"][seat] is None:
136                 print("Seat Already Empty!")

```

```

137         else:
138             print("Cancelled Ticket of",
139                   buses[bus_id]["Seats"][seat]["Name"])
140             buses[bus_id]["Seats"][seat] = None
141             print("Ticket Cancelled Successfully!")
142         except:
143             print("Invalid Input!")
144     def search_ticket():
145         name = input("Enter Passenger Name: ").lower()
146         found = False
147         for bus_id, bus in buses.items():
148             for seat, data in bus["Seats"].items():
149                 if data and data["Name"].lower() == name:
150                     line()
151                     print("PASSENGER DETAILS")
152                     line()

```



```

153         print("Passenger :", data["Name"])
154         print("Phone      :", data["Phone"])
155         print("Bus        :", bus["Bus"])
156         print("Route      :", bus["Route"])
157         print("Seat No    :", seat)
158         print("Departure  :", bus["Time"])
159         found = True
160     if not found:
161         print("Passenger Not Found!")
162 def booking_details():
163     line()
164     print("ALL BOOKINGS")
165     line()
166     empty = True
167     for bus_id, bus in buses.items():
168         print(f"\n{bus['Bus']} - {bus['Route']}")
169         for seat, data in bus["Seats"].items():
170             if data:
171                 print(f"Seat {seat:02d} -> {data['Name']}")
172             empty = False
173     if empty:
174         print("No Bookings Found!")
175     line()
176 def bus_summary():
177     line()
178     print("BUS SUMMARY REPORT")
179     line()
180     for i, b in buses.items():
181         print(f"\n{b['Bus']}")
182         print("Route      :", b["Route"])
183         print("Booked      :", booked(i))
184         print("Available   :", available(i))
185         print("Collection  : ₹", booked(i) * b["Fare"])
186     print("\nTotal Revenue : ₹", total_collection())
187     line()
188 def booking_history_view():
189     line()
190     print("BOOKING HISTORY")
191     line()
192     if not booking_history:
193         print("No Booking History!")
194     else:
195         for i in booking_history:
196             print(f"{i['Passenger']} booked Seat {i['Seat']} "
197                   f"in {i['Bus']}")
198     line()
199 while True:
200     clear()
201     line()
202     print("# Real Time Bus Ticket Booking Management System")
203     line()
204     print("1. View Available Buses")
205     print("2. View Seat Layout")

```

```

206 print("3. Book Ticket")
207 print("4. Cancel Ticket")
208 print("5. Search Passenger")
209 print("6. View All Bookings")
210 print("7. Bus Summary Report")
211 print("8. Booking History")
212 print("9. Exit")
213 line()
214 choice = input("Enter Your Choice: ")
215 clear()
216 if choice == "1":
217     show_buses()
218 elif choice == "2":
219     show_buses()
220     try:
221         bus_id = int(input("Select Bus Number: "))
222         seat_layout(bus_id)
223     except:
224         print("Invalid Input!")
225 elif choice == "3":
226     book_ticket()
227 elif choice == "4":
228     cancel_ticket()
229 elif choice == "5":
230     search_ticket()
231 elif choice == "6":
232     booking_details()
233 elif choice == "7":
234     bus_summary()
235 elif choice == "8":
236     booking_history_view()
237 elif choice == "9":
238     print("Thank You For Using Bus Booking System")
239     break
240 else:
241     print("Invalid Choice!")
242 input("\nPress Enter To Continue...")

```

Output

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 1
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
                        ONLINE BUS BOOKING SYSTEM
=====
1. Orange Travels
  Route      : Bangalore -> Hyderabad
  Bus Type   : AC Sleeper
  Departure  : 09:00 PM
  Fare       : ₹850
  Available  : 20 Seats
-----
2. VRL Travels
  Route      : Bangalore -> Chennai
  Bus Type   : Non AC Seater
  Departure  : 10:30 PM
  Fare       : ₹650
  Available  : 20 Seats
-----
3. SRS Travels
  Route      : Bangalore -> Mumbai
  Bus Type   : Volvo Multi Axle
  Departure  : 07:00 PM
  Fare       : ₹1200
  Available  : 20 Seats
-----
4. KPN Travels
  Route      : Bangalore -> Goa
  Bus Type   : AC Sleeper
  Departure  : 08:15 PM
  Fare       : ₹950
  Available  : 20 Seats
-----
Press Enter To Continue...|
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 2

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
ONLINE BUS BOOKING SYSTEM
=====
1. Orange Travels
  Route      : Bangalore -> Hyderabad
  Bus Type   : AC Sleeper
  Departure  : 09:00 PM
  Fare       : ₹850
  Available  : 20 Seats
-----
2. VRL Travels
  Route      : Bangalore -> Chennai
  Bus Type   : Non AC Seater
  Departure  : 10:30 PM
  Fare       : ₹650
  Available  : 20 Seats
-----
3. SRS Travels
  Route      : Bangalore -> Mumbai
  Bus Type   : Volvo Multi Axle
  Departure  : 07:00 PM
  Fare       : ₹1200
  Available  : 20 Seats
-----
4. KPN Travels
  Route      : Bangalore -> Goa
  Bus Type   : AC Sleeper
  Departure  : 08:15 PM
  Fare       : ₹950
  Available  : 20 Seats
-----
Select Bus Number: 1
=====
SEAT LAYOUT
=====
[01:A] [02:A] [03:A] [04:A]
[05:A] [06:A] [07:A] [08:A]
[09:A] [10:A] [11:A] [12:A]
[13:A] [14:A] [15:A] [16:A]
[17:A] [18:A] [19:A] [20:A]

A = Available | X = Booked
=====
```

```

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 3

```

Ticket Booked Successfully!

```

=====
                        BUS TICKET
=====
Booking Time : 20-05-2026 11:05 AM
Passenger    : rohith
Age          : 24
Gender       : male
Phone        : 9515536321
Bus Name     : Orange Travels
Route        : Bangalore -> Hyderabad
Bus Type     : AC Sleeper
Departure    : 09:00 PM
Seat Number  : 2
Fare         : ₹ 850
=====

```

Press Enter To Continue...

```

=====
Select Bus Number: 1
=====
SEAT LAYOUT
=====
[01:A] [02:X] [03:A] [04:A]
[05:A] [06:A] [07:A] [08:A]
[09:A] [10:A] [11:A] [12:A]
[13:A] [14:A] [15:A] [16:A]
[17:A] [18:A] [19:A] [20:A]

A = Available | X = Booked
=====
Enter Seat Number: 2
Seat Already Booked!

Press Enter To Continue...

```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 4

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
                        ONLINE BUS BOOKING SYSTEM
=====
1. Orange Travels
  Route      : Bangalore -> Hyderabad
  Bus Type   : AC Sleeper
  Departure  : 09:00 PM
  Fare       : ₹850
  Available  : 19 Seats
-----
2. VRL Travels
  Route      : Bangalore -> Chennai
  Bus Type   : Non AC Seater
  Departure  : 10:30 PM
  Fare       : ₹650
  Available  : 20 Seats
-----
3. SRS Travels
  Route      : Bangalore -> Mumbai
  Bus Type   : Volvo Multi Axle
  Departure  : 07:00 PM
  Fare       : ₹1200
  Available  : 20 Seats
-----
4. KPN Travels
  Route      : Bangalore -> Goa
  Bus Type   : AC Sleeper
  Departure  : 08:15 PM
  Fare       : ₹950
  Available  : 20 Seats
-----
Select Bus Number: 1
Enter Seat Number: 2
Cancelled Ticket of rohith
Ticket Cancelled Successfully!

Press Enter To Continue...|
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 5

Enter Passenger Name: rohith
Passenger Not Found!

Press Enter To Continue...|
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 6

=====
ALL BOOKINGS
=====

Orange Travels - Bangalore -> Hyderabad
Seat 02 -> rohith

VRL Travels - Bangalore -> Chennai

SRS Travels - Bangalore -> Mumbai

KPN Travels - Bangalore -> Goa
=====

Press Enter To Continue...|
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 7
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
BUS SUMMARY REPORT
=====

Orange Travels
Route      : Bangalore -> Hyderabad
Booked     : 1
Available  : 19
Collection : ₹ 850

VRL Travels
Route      : Bangalore -> Chennai
Booked     : 0
Available  : 20
Collection : ₹ 0

SRS Travels
Route      : Bangalore -> Mumbai
Booked     : 0
Available  : 20
Collection : ₹ 0

KPN Travels
Route      : Bangalore -> Goa
Booked     : 0
Available  : 20
Collection : ₹ 0

Total Revenue : ₹ 850
=====

Press Enter To Continue...|
```



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 8

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
BOOKING HISTORY
=====
rohith booked Seat 2 in Orange Travels
rohith booked Seat 2 in Orange Travels
rohith booked Seat 2 in Orange Travels
=====

Press Enter To Continue...|
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

=====
# Real Time Bus Ticket Booking Management System
=====
1. View Available Buses
2. View Seat Layout
3. Book Ticket
4. Cancel Ticket
5. Search Passenger
6. View All Bookings
7. Bus Summary Report
8. Booking History
9. Exit
=====
Enter Your Choice: 9
```

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

Thank You For Using Bus Booking System
PS D:\python\mini projects\rohith> |
```

Conclusion

The **Real Time Bus Ticket Booking Management System** successfully demonstrates the implementation of a computerized reservation and seat management system using Python programming. The project provides an efficient and user-friendly solution for managing bus ticket booking operations in real time. By replacing manual reservation methods with an automated system, the application helps reduce human errors, avoids duplicate seat bookings, and improves the overall efficiency of transportation management.

The system includes various useful features such as viewing available buses, checking seat availability, booking tickets, cancelling reservations, searching passenger details, generating booking summaries, and tracking revenue collections. These features make the application similar to real-world online bus reservation platforms and help users understand how modern transportation booking systems operate. The real-time seat tracking mechanism ensures accurate booking management and improves reliability within the system.

The project also provides practical exposure to important Python programming concepts such as functions, loops, dictionaries, conditional statements, exception handling, modular programming, and data handling techniques. Through the development of this application, the understanding of problem-solving, logical thinking, and system design is significantly improved. The menu-driven console interface makes the program simple, interactive, and easy to use for both beginners and end users.

Another major advantage of this project is its scalability and flexibility. Additional features such as online payment integration, database connectivity, user authentication, QR-based ticket generation, and graphical user interfaces can be added in the future to make the system more advanced and industry-oriented. This project can serve as a foundation for developing larger transportation management applications used in real-world environments.

In conclusion, the **Real Time Bus Ticket Booking Management System** is an effective mini project that combines programming knowledge with real-world application development. It simplifies the ticket reservation process, improves management efficiency, and demonstrates the practical use of Python in developing real-time management systems. The project is highly useful for academic learning and provides a strong understanding of software development concepts related to transportation and reservation systems.



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